

Project Report

ME-482: Biomechanics of the musculoskeletal system

Hydrogel Injection to Stiffen the Soft Palate and Treat Obstructive Sleep Apnea

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Abstract

Obstructive Sleep Apnea (OSA) is a common disorder characterized by repeated blockage of the upper airway during sleep, leading to partial (hypopneas) or complete (apneas) interruptions in breathing. This condition disrupts the sleep-wake cycle, resulting in fragmented sleep and reduced sleep quality. Current treatments, such as positive pressure therapy, upper airway surgery, and neurostimulation, have limitations. To address these limitations, this study proposes a new strategy involving the injection of a stiffening hydrogel into the soft palate. First, a numerical model was developed to demonstrate that hydrogel injection could reduce soft palate displacement and effectively eliminate upper airway obstructions in OSA patients. Accordingly, a new hydrogel with suitable mechanical properties was synthesized using chitosan, glycerophosphate, and alginate as main components. Overall, this study discusses the possibility of using an injectable hydrogel as a new approach to treat OSA and paves the way for future experiments on animal tissues.

1 Introduction

Obstructive sleep apnea (OSA) is a prevalent condition, affecting approximately 12% of men and 6% of women aged 40 and above in Switzerland [1]. It is characterized by repetitive collapse of the pharynx during sleep, causing a reduction in the upper airway's diameter and subsequently increasing airway resistance (Figure 1). This resistance leads to partial (hypopneas) or complete (apneas) interruptions in ventilation, resulting in respiratory acidosis, hypercapnia, and eventually awakening from sleep [2].

During apnea, the body typically attempts to

compensate for blood gas abnormalities by adjusting the respiratory effort through activation of the sympathetic nervous system. While this mechanism is efficient in regulating metabolic disturbances, it can lead to awakenings from sleep. For patients, this disruption at the sleep-wake transition often elevates blood pressure and, over time, increases the risk of heart failure and sudden death [3]. Additionally, sleep fragmentation induced by these brief periods of awakening is responsible for excessive daytime sleepiness, and neurocognitive deficits and places patients at heightened susceptibility to accidents associated with sleep deprivation [4].

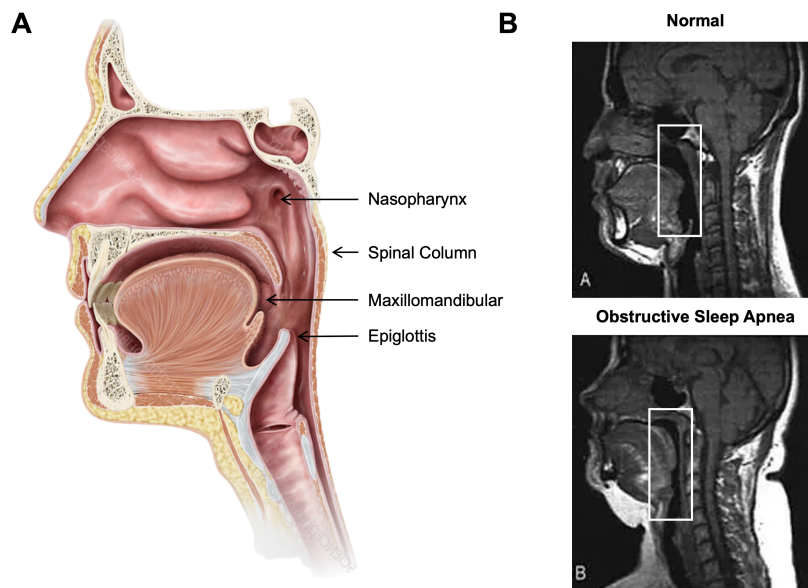


Figure 1: (A) Physiology of the upper airway. (B) MRI rendering of a patient without obstructive sleep apnea (OSA) (left panel) and a patient with OSA (right panel). The upper airway region is highlighted.¹

The interplay between dilating and collapsing forces in the upper airway region is a key driver in OSA [5]. Dilating forces primarily arise from the muscles of the oropharynx and, to a lesser extent, from the positive pressure induced in late inspiration. Conversely, collapsing forces are associated with intra-luminal upper airway pressures and the forces exerted on the pharyngeal wall [3]. During wakefulness, upper airway dilator muscles display high activity and maintain the airway open. During sleep, this muscle activity is reduced. For patients presenting risk factors such as obesity, male sex, age, menopause, fluid retention, adenotonsillar hypertrophy, or smoking, this can often lead to airway obstruction [2].

The upper airway is a flexible structure composed of varying amounts of muscular and adipose tissue, which differs between patients. Only passively supported by bony structures for the most part, the upper airway lacks rigid support, leading it to behave like a collapsible cylinder. This collapsibility is influenced by well-defined anatomical factors such as the softening of tissue, skeletal components and changes in the sensory and neural structures predisposing the upper airway to collapse [3].

In the current landscape of treating OSA (overview in Table 1), numerous options are available to assist patients; however, these treatments often fall short of complete efficacy or adoption. The prevailing method is Continuous Positive Airway Pressure (CPAP) therapy, typically prescribed for moderate to severe OSA cases. CPAP operates on the principle of delivering pressurized air to prevent airway collapse, achieved through the consistent use of a mask covering the mouth and/or nose during sleep. While CPAP doesn't offer a cure for OSA, it effectively manages the condition. Despite its proven effectiveness, approximately 40% of patients struggle to adhere to CPAP treatment in the long term. This non-compliance is attributed to various factors including mask-related discomfort (skin irritation/allergies, claustrophobia, and mask leak), pressure-related complications (headaches, sinusitis, and rhinitis), and equipment-related issues (noise, and odour) [6].

Other pressure-based therapies include the

application of bi-level positive airway pressure (BiLevel), where higher pressure is applied during inhalation than exhalation. While randomized trials have shown no major benefits compared to CPAP, BiLevel is sometimes preferred by patients who experience discomfort with expiratory pressure. Similarly, auto-titration positive airway pressure has been used to vary pressure on an ad-hoc basis to maintain stable ventilation based on body posture or sleep stage. Once again, most randomized trials have shown no improvement in adherence with auto-titration compared with standard continuous positive airway pressure [2].

For patients who do not respond well to positive airway pressure, alternative options include oral devices, upper airway surgery and positional therapy. Oral devices for the treatment of OSA generally function by applying pressure to the jaw to prevent the posterior movement of the tongue to the back of the upper airway (retroglottal collapse). While they are preferable for patients with mild-to-moderate disease, the efficacy of oral devices varies and little data on outcomes have been collected [2]. Standard procedures such as uvulopalatopharyngoplasty, involving the removal of excess tissue in the throat to widen the airway, can be performed on patients suffering from OSA. However, improvements in symptoms are generally small, with fewer than 50% of patients experiencing major improvements in the apnoea-hypopnoea index² [2]. Other surgeries such as maxillomandibular advancement (MMA), where the upper and lower jaw bones are repositioned to increase the tension on the pharyngeal soft tissues and expand the airway, are likely more effective than simple procedures. However, MMA is an invasive procedure that can alter facial appearance and requires weeks of recovery [7].

Other strategies involve using electrodes to stimulate muscles and widen the oropharynx radius during sleep. For example, Inspire[®] therapy consists of an implant that stimulates glossal muscles to maintain the tongue in an anterior position, thereby freeing the upper airway in OSA patients [8]. Activated only while sleeping, this device is synchronized with respiration to ensure optimal results. While this therapy has proven effective in 79% of cases [9], the surgical

¹Retrieved and adapted from SCIENCEPHOTOLIBRARY and Wickramasinghe, H. (n.d.). Obstructive Sleep Apnea (OSA) Workup. In H. Wickramasinghe (Ed.), Medscape.

²Measure used in sleep medicine to quantify the severity of sleep apnea.

Table 1: Pros and cons of current therapies for OSA

	Pros	Cons
Positive Airway Pressure & BiLevel	Effectively manages the condition	Poor treatment adherence
Oral Devices	Non-invasive and convenient	Questionable efficacy
Uvulopalatopharyngoplasty	Can be very efficient	Poor success rate
Maxillomandibular Advancement	Can be very efficient	Highly invasive
Electrode Stimulation	Efficient most of the time	Limited patient adoption

procedure required to place the implant remains a barrier to patient adoption.

Conservative measures such as reducing alcohol intake or avoiding the supine position (lying horizontally with the face and torso facing up) can enhance the effectiveness of therapies. Additionally, weight loss and neuromuscular exercises are considered beneficial, even though current data are inconclusive [2]. Unfortunately, despite efforts, these measures rarely succeed in increasing patient adherence to existing treatments. Consequently, there is still a pressing need for the development of a convenient and minimally invasive treatment for OSA.

In this context, the application of an injectable hydrogel (3D water-swollen polymer networks that mimic the properties of many tissues [10]) to stiffen oro-pharyngeal tissues emerges as a promising treatment for OSA. This approach presents notable advantages, including the ease of administration through simple injections, minimal morbidity due to its non-surgical nature, and reversibility. Administered under local anaesthesia, this procedure would be accessible, non-invasive and cost-effective. Recognizing the pivotal role of patient adherence in OSA management, the introduction of injectable hydrogel underscores the significance of patient comfort and compliance in achieving successful outcomes.

Here, we present a novel approach for the treatment of OSA in patients who are not compatible with the already existing ones or patients who would like a more convenient treatment. In collaboration with Dr. Karma Lambercy³ and inspired by Ece Uslu’s⁴ work on Tracheomalacia [11], we developed a minimally in-

vasive therapeutic approach involving the injection of hydrogel in the soft palate to stiffen the tissue and widen the upper air-way of patients suffering from OSA. After defining key engineering requirements, we screened for hydrogel candidates and eventually designed our own composition. A numerical model was developed to test the hypothesis that internal mechanical support provided by the hydrogel could stiffen the soft palate and prevent airway obstruction. The model demonstrated that injecting a hydrogel could help compensate for muscle relaxation by restricting the soft palate displacement. Based on these findings, a new injectable hydrogel was formulated using chitosan, glycerolphosphate and alginate as main components. This hydrogel, whose mechanical properties were tested and validated for injectability and sufficient stiffness, emerges as a promising candidate for treating OSA.

2 Proposed Hydrogel-Based Treatment

2.1 Seminal Work on Tracheomalacia

Uslu et al. (2023) proposed a proof-of-concept study demonstrating the potential use of hydrogel to stiffen the airway in the context of Tracheomalacia (TM), a condition characterized by weak tissues leading to tracheal collapse in newborns [11]. The team introduced a new treatment strategy aimed at mechanically supporting the trachea by wrapping an adhesive hydrogel patch around it (extraluminally). They envisioned that adhesive hydrogels could address several surgical challenges associated with treating TM, such as the trauma caused by extralumi-

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nal stent fixation sutures to surrounding tissues. Additionally, they anticipated that the softness of hydrogels could provide necessary flexibility to the trachea, reducing potential pain during natural movement compared to stiffer materials. Through a numerical model, the team demonstrated that external mechanical support provided by a hydrogel patch could prevent airway collapse. Based on these findings, they synthesized a new hydrogel capable of robust adhesion on wet tracheal surfaces, utilizing hydroxyethyl acrylamide (HEAam) and polyethylene glycol methacrylate (PEGDMA) as the main polymer network and crosslinker, respectively. Finally, their *ex vivo* experiments revealed that adhesive hydrogel patches could effectively restrain airway collapse under negative pressure.

Inspired by the work of Uslu et al. on Tracheomalacia (TM), this report proposes a new treatment strategy aimed at mechanically supporting the oropharynx as a potential alternative treatment for OSA. Compared to the trachea, targeting the upper airway region poses specific challenges that must be considered: (1) Addressing the dynamic nature of the soft tissues in the oropharyngeal region, particularly those involved in speaking and swallowing; (2) Overcoming the issue of the upper airway not being a closed tube, thereby rendering conventional wrapping of a patch around it impossible; and (3) Ensuring treatment reversibility to allow adjustments or removal if complications arise, especially considering the complex functions and delicate structures of the oropharyngeal tissues.

2.2 Soft Palate as Therapeutic Target

The oropharynx comprises the soft tissue area delineated by the nasopharynx as the upper limit, the epiglottis as the lower limit, the maxillo-mandibular complex anteriorly, and the spinal column posteriorly [12]. Within this region, the muscle activities of the soft palate, tongue, and pharyngeal wall are critical to maintaining pharyngeal airway patency for proper ventilation and gas exchange. Pharyngeal muscles counteract negative transluminal air pressure that tends to collapse the airway during inspiration. In obstructive sleep apnea, the activities of pharyngeal muscle activities are suppressed during sleep, allowing the pharyngeal airway to collapse

during inspiration [13].

The genioglossus is the largest upper airway dilator muscle. Its contraction results in anterior movement of the tongue and widening of the oropharyngeal airway. In normal settings, this muscle is tonically active to maintain the tongue position and is phase active to counteract intraluminal negative pressure during inspiration.[13] Several studies have demonstrated the importance of genioglossus relaxation in OSA and have shown that treatments such as hypoglossal nerve stimulation or muscle training exercises might be useful.[2] However, the intrinsic complexity of normal tongue movement in activities such as speaking or swallowing makes it hard to target for mechanical stiffening and poses challenges for patient compliance, particularly due to comfort.

The soft palate plays a crucial role in directing the flow of air and ensuring that ingested food is diverted away from the nasal cavity. Composed of muscles and fat tissue, it extends posteriorly from the hard palate and acts as a valve. During nasal breathing, it depresses to seal the oropharyngeal isthmus, while it elevates during oral breathing or swallowing to separate the nasopharynx from the oropharynx by closing the pharyngeal isthmus[12] (see Figure 2).

The primary muscles responsible for controlling palatal position are the levator veli palatini and the palatoglossus. Both muscles are active during oral and nasal breathing, but the levator veli palatini is more active during oral breathing, whereas the palatoglossus predominates during nasal breathing [13]. Anderson et al. [14] analyzed the contribution of intrinsic and extrinsic portions of the soft palate muscles in velopharyngeal and oropharyngeal constriction in a 3D modelling study.

In sleeping OSA patients, abnormal muscle relaxation and tissue flexibility due to fat accumulation may disrupt the proper function of the palatal valve and induce partial or complete disruption of ventilation. Therefore, stiffening the soft palate offers a minimally invasive approach that may restore its function as an effective valve, promoting better airflow control and reducing the risk of airway obstruction. This underscores its potential as a viable option for individuals with OSA seeking effective and well-tolerated therapeutic alternatives.

⁵Adapted from Drake, R. L., Vogl, W., Mitchell, A. W. M., & Gray, H. (2020). *Gray's anatomy for students* (4th ed.). Elsevier. p. 1076, 1088 and 1098.

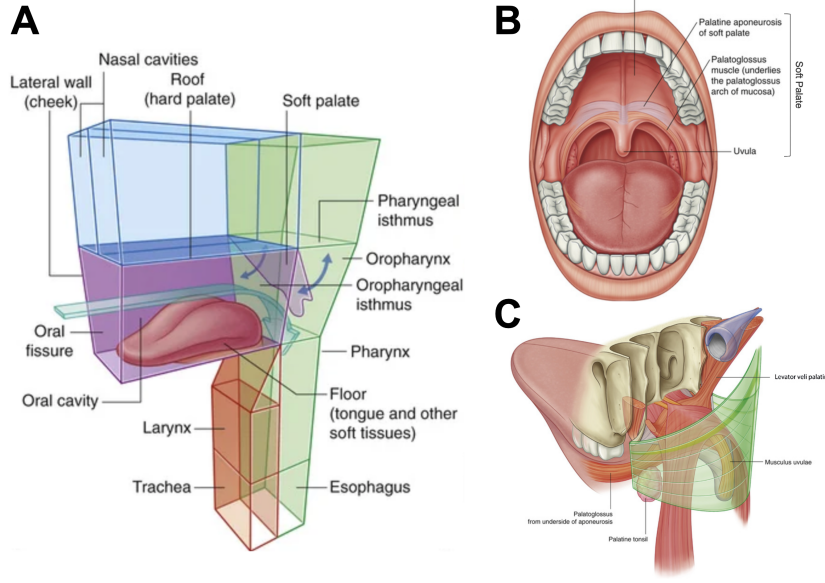


Figure 2: (A) Representation of the pharyngeal cavity (green) and its relationship with other cavities influenced by the valve-like nature of the soft palate. (B) Anatomy of the soft palate viewed from the inside of the mouth. (C) Posterior view of the primary muscles responsible for controlling palatal position (levator veli palatini and palatoglossus).⁵

2.3 Mechanical Properties of the Soft Palate

Understanding the mechanical properties of the soft palate is essential to optimize the hydrogel stiffening procedure. In this section, we extracted several mechanical properties of the palatal tissues from the literature. The final properties considered for the remainder of this study are displayed in Table 2.

Birch et al. [15] conducted an ex-vivo study of the biomechanical properties of the human soft palate based on 10 palates harvested from normal adult human cadavers. Taking uniaxial measurements at different zones between the uvula and hard palate, the team described Young’s modulus (E) values ranging from 0.58 ± 0.28 kPa to 1.41 ± 0.68 kPa, showing a non-uniform distribution across the palate that increases toward the hard palate. Intuitively, these values seem rather low compared to the reference mechanical properties of muscles and fat tissues in-vivo (muscles: $E = 10$ kPa and fat tissues: 2 kPa [16]). This could be explained by the fact that samples were collected from cadavers, which is not optimal for determining the in-vivo mechanical properties of a given tissue. Similarly, retrieved Poisson’s ratio values ranged from 0.30 ± 0.21 to 0.45 ± 0.26 .

Haddad et al. [17] determined the Young’s modulus of the soft palate based on indentation testing of fresh human OSA specimens obtained during surgery. Their results showed a Young’s modulus of 6.54 ± 3.52 kPa, which was corroborated by Brown et al. [18] measurements using Magnetic Resonance Elastography in-vivo. Part of the relatively large difference between the two measurements could be attributed to the rigor mortis effects known to alter the mechanical properties of tissues in cadavers. Moreover, the uniaxial testing method adopted by Birch et al. requires long cylindrical probes, which are challenging to properly extract ex-vivo and make the measurement prone to inaccuracy errors.

Overall, the Young’s modulus values reported by Haddad et al. and Brown et al. seem more plausible based on the reference values for muscle and fat tissues and will be therefore considered in this study.

Finally, Subramaniam et al. [19] conducted a study on 2 female subjects with OSA, which may serve as a valuable reference point for our numerical simulations and further ex-vivo experiments. Using in-vivo MR imaging technology, the team described a change in velopharyngeal resistance of 60%, in maximal stress of 167%, and in maximum displacement of 400% from healthy to OSA

Table 2: Mechanical properties of the soft palate from the literature

Reference	Method	Status	Young's modulus	Poisson's ratio
Birch et al. [15]	Uniaxial test ex-vivo on cadaver tissues	Normal	0.58 kPa to 1.41 kPa	0.30-0.45
Haddad et al. [17]	Indentation testing of fresh human specimens	OSA	6.54 kPa	-
Brown et al. [18]	Magnetic Resonance Elastography in-vivo	OSA	7.26 kPa	-
Selected	-	Normal & OSA	6.54 kPa	0.45

patients. (Table 3) However, it's important to note that the sample size of two patients may limit the generalizability and relevance of the findings to a broader population.

Table 3: Mechanical measurements on normal vs OSA soft palate [19]

	Control	OSA
Maximum flow velocity [m/s]	2.4	2.5
Velopharyngeal Resistance [Pa/lpm]	0.5	0.3
Maximum Stress [Pa]	240	400
Maximum Displacement [mm]	0.4	1.6

2.4 Injection Procedure

This section describes in detail the hydrogel injection procedure designed under the supervision of Dr. Karma Lamercy, based on post-mortem observations (Figure 3A) and live endoscopy on an OSA patient. Dr. Karma Lamercy's insights were extremely useful in developing a procedure that ensures precise delivery of the hydrogel material while minimizing patient discomfort and maximizing efficacy.

The proposed procedure consists of multiple injections of the hydrogel agent in a liquid state beneath the lining tissue of the soft palate. Administered directly through the oral cavity, these injections are conducted in-office under topical anaesthetic spray to ensure patient comfort.

Each injection, typically penetrating less than 2 mm in depth and covering an approximate area of 2 cm², is placed between the fat and muscle tissues of the palate to optimize outcomes and ensure proper physiological function-

ality carried out by the muscles. The soft palate is arbitrarily divided into four distinct injection zones of approximately 2 cm² each. (Figure 3) The same amount of hydrogel will be injected into the middle of each zone to ensure proper distribution of the viscous hydrogel in the tissues. Upon injection, the hydrogel should polymerize after approximately 60 seconds.

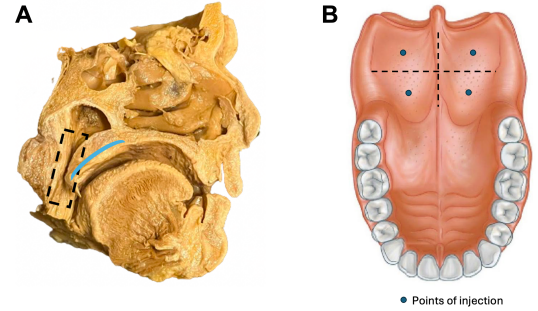


Figure 3: (A) Representation of the airway collapse (black dashed rectangle) and the injection zone (blue line) on the picture of a human post-mortem face taken at the CHUV. (B) Division of the soft palate in four injection zones.⁶

The hydrogel of interest is expected to degrade significantly in the first few days following the injections before stabilizing after approximately one week. To anticipate the volume lost due to degradation, we propose to reinject four new doses a few days after the initial injection round. Overdosing the first injection was also considered; however, this could lead to serious discomfort for the patient. Additionally, through the second round of injections, it should be easier for the practitioner to precisely adjust the amount of hydrogel necessary to induce the desired stiffening effect.

The hydrogel used for this procedure is de-

⁶Adapted from Drake, R. L., Vogl, W., Mitchell, A. W. M., & Gray, H. (2020). Gray's anatomy for students (4th ed.). Elsevier. p. 1076, 1088 and 1098.

signed to gelify almost instantly after mixing its polymer network and crosslinker components. With a traditional single-channel syringe, the practitioner needs to mix the two components before injecting the mixture into the patient's tissues. To make the procedure easier, we recommend using a syringe with two channels. With this setup, the practitioner can load the two components independently into the syringe, and mixing happens directly in the tip of the syringe. As the hydrogel flows in a liquid state through the needle into the tissues, it starts to gelify. Once in the tissues, the mixture takes a few seconds to achieve its final state.

3 Hydrogel Formulation

3.1 Engineering Requirements

In this study, we will focus our research on an injectable hydrogel. Indeed, the injectability is an important characteristic of the material as it renders it possible to implant the hydrogel into living bodies in a minimally invasive way. In addition, it requires only standard equipment, most of the time a simple syringe is sufficient. Injectable hydrogels also have the advantage of being able to conform to the shape of tissue or organ being treated due to its fluid state during injection. This technology has a huge potential in various biomedical applications. In order to select the most suitable injectable hydrogel to stiffen the soft palate, a special focus on four different properties is made.

First, biocompatibility is necessary for biomedical applications. The biocompatibility of the base composition of the hydrogel should be confirmed as well as the degraded product after it has been injected into the living body. Second, mechanical properties are important in order to stiffen the soft palate. The goal is to get a similar strength between a patient suffering from OSA and a healthy patient. This property can be easily tuned by modifying the composition of the hydrogel, for example by introducing different functional groups or increasing the degree of cross-linking. The properties of the hydrogel must be compatible with an injection targeting the space between muscles and fat tissues. Third, the gelation technique is a non-negligible parameter as the hydrogel will be injected inside human tissues. The aim is therefore to elaborate a composition without a light-based or UV-

based curing. Material and chemical knowledge grant to use another method to create a gel. The gel state can also be reversible, also called thermoreversible, to offer the possibility to remove the hydrogel injected if the patient is uncomfortable with it. The thermo-responsiveness of the hydrogel relies mainly on T_g and the temperature at which the hydrogel is operating. The goal in our case is to obtain a T_g between room temperature and 37°C in order to allow the solution to be fluid, thus injectable at room temperature, and to gelify once it has been injected in the soft palate of the patient. The reversibility will be obtained by cooling the hydrogel below T_g , by applying ice for example, allowing the hydrogel to fluidify and be removed with a syringe. It has to be mentioned that while sleeping, the temperature of the body is constant, and thus the mechanical properties of the hydrogel can be maintained during the night. And lastly, the biodegradability of the hydrogel is wished to last only 3 to 6 months regarding the needs described by our medical advisor Dr. Karma Lambercy [20].

3.2 Candidate Formulations

Through a literature review, we identified two compositions with promising mechanical properties described below in two distinct papers by Pang et al. and Nair et al. respectively. These properties will be thoroughly evaluated and rated in Tables 4 and 5. In these tables, a minus (-) will indicate a negative contribution to the property for the studied hydrogel, while a plus (+) will indicate a positive contribution for both of the hydrogels under study.

Composition 1 proposed by Pang et al. (2020) [21]

The first candidate for this study is composed of Carboxymethyl Chitosan (CMCS) with chitin nano-whiskers (CtNWs). The main advantage of this hydrogel is the tunability of its composition. Indeed, the Young's modulus and the gelation time can be modified by simply changing the concentration of the CtNWs and DDA respectively. By simply varying the wt % of CtNWs between 0 and 1 (refers as CD0, CD0.5 or CD1), values of Young's modulus between approximately 40 and 70 kPa can be reached. This is already one order of magnitude above the range of Young's modulus of soft tissues (1.5-7

Table 4: Properties of Composition 1 [21]

	Paper 1
Composition	Chitosan at a deacetylation of 85% and Mw=1400 kDa to create Carboxymethyl chitosan (CMCS) with chloroacetic acid and Dextran T-500 at Mw = 500 kDa oxidized by periodate to generate Dextran Dialdehyde (DDA)
Injectability	+++ with double-tube syringe
Cytocompatibility	+++
Biodegradability	++ (50% in 1 day and then stabilize around 40% after 10 days)
Gelation method	Chemical reaction (via Schiff's base reaction)
Gelation time	Tunable with DDA concentration and its oxidation degree (7-30s)
Young's modulus	CD0= 43.2±4.6 kPa, CD0.5= 47.2±5.6 kPa, CD1=67±6.2kPa
Swelling performance	CD0 after 36 h was 52.62 ± 8.59%
Reversibility	- -
Antibacterial	++ with sufficient inhibition of bacterial growth to protect wounds

Table 5: Properties of Composition 2 [22, 23]

	Paper 2
Composition	Chitosan at a deacetylation > 80% and Mw ≈ 300 kDa (200mg), HCl at 0.1M (9mL), Glycerol phosphate disodium salt (560 mg)
Injectability	+++ (Each injection was 0.2 ml in volume and performed through hypodermic syringe equipped with a gauge 22G1 needle Novel injectable neutral solutions of chitosan form biodegradable gels in situ paper)
Cytocompatibility	+++
Biodegradability	++ (30% in 1 day and then stabilize around 55% after 25 days)
Gelation method	Thermogelling at physiological pH between Chitosan solution and Glycerol phosphate disodium salt
Gelation time	Few seconds
Young's modulus	No info only G'=3000 Pa at 37°C with Tg around 36°C
Swelling performance	≈ 2
Reversibility	+++ (Thermoreversibility)
Antibacterial	No info

kPa) [17].

Another key factor is the gelation method of the hydrogel. The gelation is a chemical reaction (Schiff's base reaction) and starts when the components are mixed. The change of state occurs after 30 seconds. This gelation method is attractive as it does not require any external energy such as UV light. The use of UV light creates probable issues when working with human tissues, thus it should be avoided. The overall phase transformation is fast and, in the same time, lets enough time to the solution to flow and gel rapidly into the tissues.

Then, the biodegradability has also been taken into account as the solution will last for few months. The paper describes a hydrogel which degrades consequently during first days (50%) and then stabilize around 40% after 10 days. In addition, tested compositions tend to shrink and not to swell like most of hydrogels. These two aspects have to be considered together to define the adequate amount needed for injection.

Regarding the cytocompatibility, there is no issue with this composition. The presence of the hydrogel in the tissues does not lead to a higher percentage of dead cells. In addition, it can also have an antibacterial effect which decrease the risk of potential infection.

Furthermore, the composition can be easily adapted to the need of each patient. This would potentially lead to a higher degree of success of the treatment.

Finally, this is an effective way to stiffen the tissues but further investigation are required to verify that it does not stiffen the injected area too much. The motivation is to avoid discomfort for the patient by deteriorating some physiological functions. Even if three different concentrations of CtNWs are reported, the hydrogel without CtNWs (CD0) shows suitable mechanical properties for our purpose (Table 4). It has a sufficient Young's modulus (43.2±4.6 kPa) supported by a simpler composition which simplifies its synthesis.

Composition 2 proposed by Nair et al. (2011) based on the work of Chenite et al. (2000) [22, 23]

These papers describe a biodegradable, non-toxic neutral injectable hydrogel based on unmodified chitosan, and Glycerol Phosphate (GP) which can be formed in situ, with no toxic cross linking chemicals involved. Chitosan is a polyside composed of a random distribution of β -(1-4)-linked D-glucosamine (deacetylated unit) and N-acetyl-D-glucosamine (acetylated unit). It is produced by chemical (in an alkaline medium) or enzymatic deacetylation of chitin, the component of the exoskeleton of arthropods (crustaceans) or the endoskeleton of cephalopods (squid, etc.) or even the wall of fungi. The two main parameters of chitosan are the molecular weight Mw and the deacetylation degree. Chitosan is mixed with HCl to obtain a

liquid solution below room temperature. In addition, the pH values range within a physiologically acceptable neutral range from 6.8 to 7.2 [22]. Nair et al. have synthesised different compositions based on Chitosan-HCl and GP-deionised water solutions. Regarding porosity and mechanical properties results [22], the most appropriate composition for OSA purpose is the one called chitosan/glycerol-phosphate C/GP based on the paper [23] as explained in Table 5.

At low temperatures, strong chitosan–water interactions protect the chitosan chains against aggregation. Upon heating, sheaths of water molecules are removed by the glycerol, which in turn allows association of chitosan macromolecules. Chitosan is neutralized thanks to the GP resulting in instantaneous gelling. In other words, GP is trapped inside the cross linked network and is bound by ionic interactions. These nearly neutral C/GP aqueous solutions will gel even faster when the temperature increases [23].

With the goal of creating a gel at physiological temperature after injection inside the human body by chemical reactions, the glass transition temperature needs to be adjusted to be lower than 37°C. In order to decrease the glass transition temperature, T_g , the paper shows that the deacetylation degree of chitosan (up to 90% [23]) needs to be increased. Molecular weight of chitosan, on the other hand, has no impact on T_g . The process to create the gel is called thermogelation. This allows the hydrogel to exhibit thermoreversible behavior as depicted by rheological tests [23]. The storage modulus G' is between 3000 Pa and 4000 Pa when the temperature is raised up to 37°C. When the hydrogel is cooled, G' decreases again with a hysteresis behavior which shows the thermoreversibility of the gelation. The hydrogel biodegradation is also detailed thanks to injection in adult Sprague–Dawley rats. Again, the degree of deacetylation was the key factor governing both the rate of degradation and the inflammatory response [23].

3.3 Selected Composition

Inspired by the two compositions screened from the literature, this section aims to present the final hydrogel composition that will be synthesized and investigated for its potential to stiffen the soft palate in the treatment of OSA. Properties of the candidates and the final formulation

are presented in Table 6.

Table 6: Summary of hydrogels' compositions

	Paper 1	Paper 2
Synthesis feasibility	+	+++
Injectability	+++	+++
Cytocompatibility	+++	+++
Biodegradability	++	++
Gelation method	Chemical reaction via Schiff's base reaction	Self thermogelation
Gelation time	+++	+++
Young's modulus	++	++
Swelling ratio	++	++
Antibacterial	++	No info

The biodegradability and the biocompatibility must be optimised. Nevertheless, regarding the lack of synthesis equipment and the ease of production for a real application, the composition C/GP from Chenite et al. [23] is chosen as the basis for further test in the laboratory.

Regarding the mechanical properties of the chosen hydrogel, material scientist's knowledge need to be applied to improve the selected composition. The expected modulus to stiffen the soft palate is three to four times the one given for the original C/GP. Thus, it was decided to add alginate to the basis composition to improve the mechanical properties while maintaining appropriate biocompatibility. The main problem with the addition of alginate is that this component does not dissolve well in water, thus it is necessary to dissolve some $CaCl_2$ into deionized water before dissolving the alginate in the solution. By doing so, it is possible to obtain an homogeneous solution which will simplify its mixing to the C/GP solution afterwards. The ions Ca^{2+} are thus the mains actors in the dissolution of alginate.

Many different compositions based on different ratio of alginic acid sodium salt, $CaCl_2$ and chitosan, inspired by Tang et al.[24] have been investigated in the lab. The final chosen composition showing promising results is made with a mass fraction (wt.%) of chitosan to alginic acid sodium salt of 1:3 and a mass ratio (wt.%) of $CaCl_2$ to alginic acid sodium salt of 21.6. With these ratios, we were able to obtain the most homogenous hydrogel and the best properties based on a first visual estimation. On the other hand, the basis composition has been taken as presented by Chenite et al. [23] in order not to include too many unknown variables.

4 Results and Discussion

4.1 Numerical Simulation

A finite element analysis (FEA) model was developed to reinforce the hypothesis that local injection in the soft palate can hinder upper airway collapse and to understand the deformation and flow response of the oropharynx under normal and OSA conditions. Since the palatal system has complex geometry and in-vivo measurements are usually not practical, we simplified the geometry to facilitate the analysis of gravitational and airflow effects. The initial planar geometry was derived from anatomical dimensions of the soft palate sourced from the literature (see Table 7).

Subsequently, the 3D model was designed using COMSOL's CAD module, while all numerical simulations were conducted in COMSOL Multiphysics. The soft palate is modeled as a rectangular thick plate with a triangular extension smoothly transitioning to a quasi-circular uvula positioned at the midpoint of its lateral edges (Figure 4). The posterior and the two lateral edges of the soft palate were considered clamped. Additionally, the flat structure is bent to a certain angle to accurately replicate physiological observations made on cadaveric tissues at the CHUV Hospital.

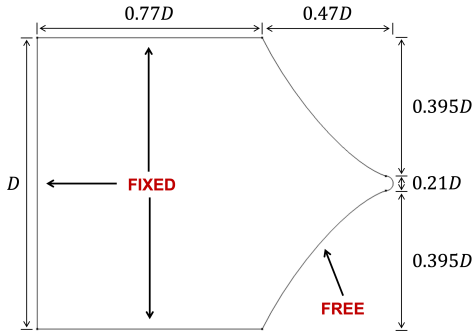


Figure 4: Model geometry for the soft palate with dimensions and the associated boundary conditions.

Throughout the simulations, sleeping patients were modeled in the supine position, with gravity simulated to act negatively on the z-axis. Three conditions were considered: (1) normal, (2) OSA (untreated), and (3) OSA treated by hydrogel injection. In the normal condition, it was assumed that muscle tension fully compensated for the effect of gravity. Conversely, in OSA patients, muscle tension was considered null. While this may not precisely mirror reality, the level

of muscle involvement in palatal displacement is challenging to assess in vivo. Thus, we estimated that these extreme conditions were sufficient for this simplified simulation. Hydrogel treatment was modeled by placing a thick rectangular plate with a triangular edge in the middle of the soft palate with a corresponding total contact surface of 14.4 cm². Attachment conditions between the soft palate tissue and the hydrogel were considered flexible.

The material constituting the soft palate was assumed to be linear, elastic and isotropic. The mechanical properties used to simulate the soft palate are listed in Table 8.

Table 8: Mechanical properties of the materials used to simulate the soft palate and hydrogel

	E	ν	ρ
Palate	6.5 [kPa]	0.45	1060 [kg/m ³]
Gel	60 [kPa]	0.49	1000 [kg/m ³]

The results of deformation induced by gravity alone are shown in Fig. 5. As expected, the maximum deformation in OSA patients was experienced at the middle of the anterior free edge. We further simulated an OSA soft palate after injection of hydrogel under identical conditions. The resulting hydrogel injection was modeled as a homogeneous material with a Young's modulus (E) equal to 60 kPa and a thickness of 0.16 cm (equal to 40% of the total thickness of the soft palate). As displayed in Figures 5B and 5C, the maximal deformation was reduced from 1.60 cm to 1.08 cm in the treated palate, corresponding to a decrease in displacement of approximately 0.5 cm. This strongly suggests that the hydrogel injection has the potential to significantly reduce upper airway obstruction by the soft palate and therefore reduce collapse in OSA patients.

We decided to push the simulations further to better understand how airflow could be impacted by the displacement of the soft palate. For this, we considered a sleeping OSA patient in a supine position exhaling through the nose. The corresponding flow chamber was modeled with rigid walls and arbitrary geometry inspired by observations made on cadaveric tissues. The airflow was modeled as incompressible and turbulent at a maximal expiration flow rate of 800 [cm³/s].[30] The fluid-structure interaction was fully coupled, indicating that changes in either the fluid or the solid could affect each other.

Table 7: Anatomical dimensions of the soft palate used for the numerical simulation [25]

	Values from Literature	Values Employed
Pharyngeal radius, D (cm)	3.80 ± 0.35 [26]	4.00 (=1.00D)
Soft palate length (cm)	4.70 ± 0.70 [27]	4.96 (=1.24D)
Palate thickness (cm)	0.37 ± 0.07 [28]	0.40 (=0.10D)
Uvula length (cm)	1.80 ± 0.10 [29]	1.88 (=0.47D)
Uvula width (cm)	0.80 ± 0.04 [29]	0.84 (=0.21D)

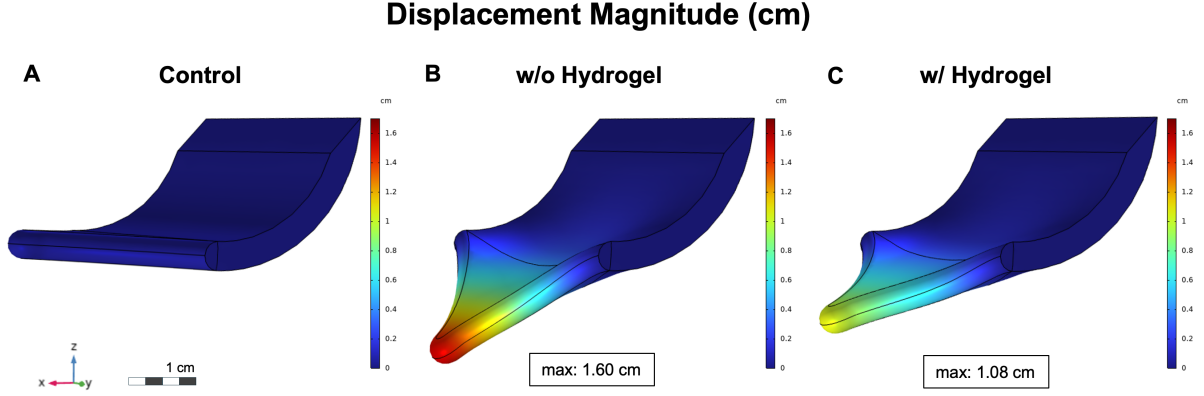


Figure 5: Soft palate deformation in (A) normal, (B) untreated OSA, and (C) treated OSA patients with hydrogel injection under gravity. Maximal displacement values are shown in black rectangles.

To achieve mesh-independent results, a parametric study of multiple refined steps was conducted. While mesh deformation in the solid domain does not cause numerical issues, the strong deformation of the fluid domain under the uvula leads to significant mesh non-orthogonalities that cause the simulation to fail at a certain refinement level. This is circumvented by solving in the "coarsest" configuration and re-meshing the deformed configuration with a finer mesh. This gives four refinement levels: "coarsest," "coarser," "coarse," and "normal," where "coarsest" is initially user-defined and the latter three are default physics-based meshes that uses the "coarsest" deformation as input. The mesh study and the corresponding pressure and velocity fields can be found in the appendix 5.

For faster-converging results, the airflow is set to laminar in the mesh study. In the real study cases, the "normal" refinement mesh option is chosen, and flow is set to turbulent as the Reynolds number exceeds values of 3000 in substantial parts of the fluid domain. A $k-\omega$ -SST model was considered for its good modeling capability in free- and wall-bounded flow. However, it was abandoned for the simpler $k-\epsilon$ -model in favor of numerically better-converging results,

making a finer mesh and a more accurate solution possible. The automatic wall treatment of COMSOL is used in the boundary layer.

Qualitative results displayed in Figure 6 demonstrate how palatal deformation, influenced by gravity and respiration, affects flow pressure and velocity at maximal expiration rate for untreated versus treated patient. In both cases, the airway contributes to approximately 2% of the deformation without a notable difference between the two scenarios. We can observe that the untreated patient shows a virtual blockage in the oro-pharynx induced by the soft palate. When the hydrogel is injected, the soft palate's displacement decreases by approximately 23%, making this obstruction less noticeable and diminishing the flow resistance. This further suggests that the injection of hydrogel with proper mechanical properties could efficiently stiffen the soft palate and significantly reduce upper airway obstruction in OSA.

4.2 Hydrogel synthesis

The components used to produce the hydrogels are chitosan (high molecular weight 310-375 kDa, $\geq 75\%$ deacetylation degree), glycerol phosphate pentahydrate (GP), alginic acid

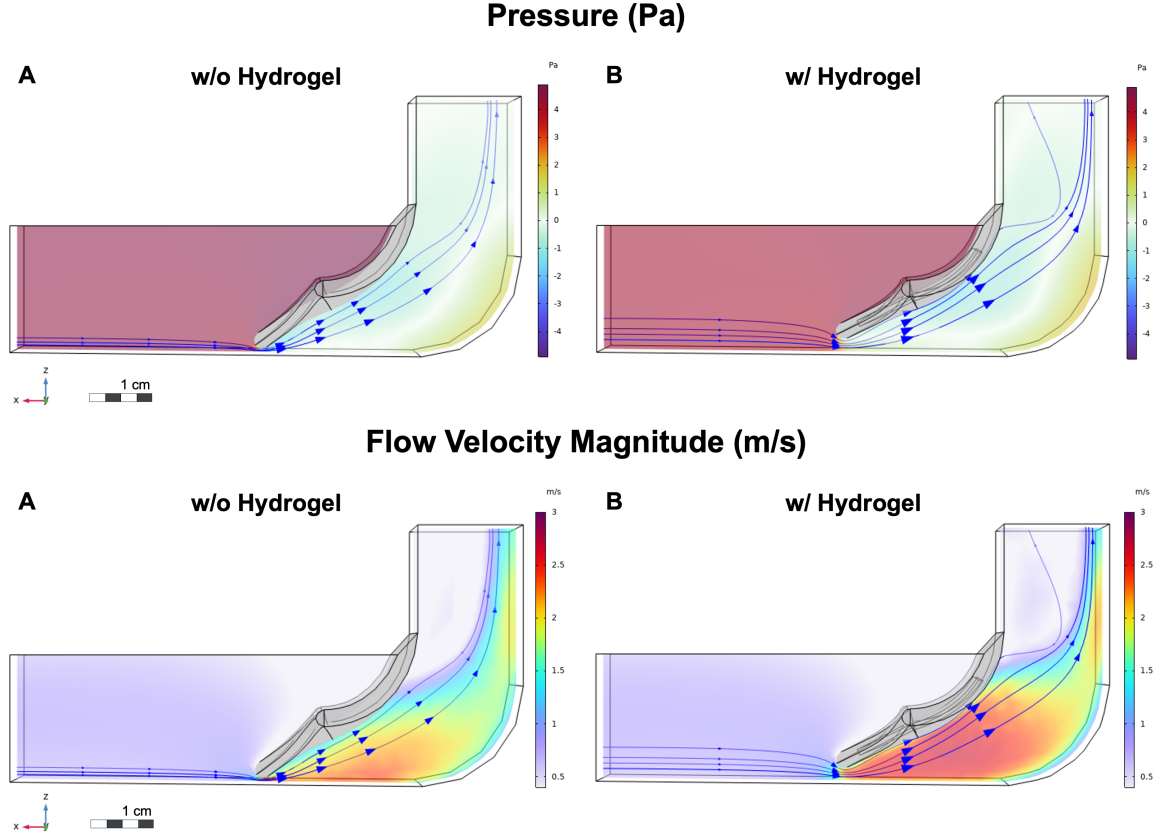


Figure 6: Pressure and velocity for (A) untreated OSA patient and (B) OSA patient treated with hydrogel under maximal expiration rate at the center of the flow chamber. Convergent velocity streamlines are represented in blue with associated arrows proportional to the flow velocity intensity.

sodium salt, calcium chloride and hydrochloric acid (0.1 M) all provided by Sigma-Aldrich®. The amount of each component is available in Table 9.

The different steps to synthesize the hydrogel are described in the following. First, the chitosan must be dissolved in the hydrochloric acid solution at room temperature under stirring. An optimal dissolution is obtained when waiting overnight. The GP is dispersed into 0.767 ml of deionized water using a vortex stirrer. Then, the $CaCl_2$ is dissolved into 1.8 ml of deionized water. Once it has been done, the alginic acid sodium salt is added to this solution gradually. The next step is to mix the obtained solution with the GP solution. And finally, the solution made of GP, alginic acid sodium salt and $CaCl_2$ is added drop by drop to the chitosan solution.

It will start to form a gel as soon as they are in contact, and the properties will build up when the gel is heated up until reaching 37°C (physiological temperature).

Table 9: Composition of the hydrogel

Component	Quantity
Chitosan	200 mg
HCl (0.1 M)	9 ml
GP	793 mg
Alginic acid sodium salt	600 mg
$CaCl_2$	130 mg

4.3 Hydrogel Testing

4.3.1 Rheological test on hydrogel

The goal of the rheological test is to evaluate the module of the complex shear modulus $G = G' + iG''$ and also the Young's modulus E . Equations 1 and 2 allow to determine the desired parameters of the samples.

$$|G| = \sqrt{G'^2 + G''^2} \quad (1)$$

$$E = 2G(1 + \nu) \quad (2)$$

Regarding the literature review, the Poisson's ratio ν is set to an average value 0.35. Thanks to the Anton Paar MCR102 modular compact rheometer, the samples (thickness of 3mm and diameter of 25 mm) can be tested (Appendix 1). The first step of these tests is to determine the amplitude range when the hydrogel exhibits a linear viscoelastic behavior. To achieve this, two amplitude sweeps are performed on two different samples. The results are presented in Appendix 2. Then, a frequency sweep is realised for two samples to estimate the storage modulus and the loss modulus of the hydrogel. The results are shown in Figure 7.

The average storage modulus G' is 18801.9 Pa and the average loss modulus G'' is 3232.3 Pa at 37°C. Thanks to previous formulas, the shear modulus G is 19077.7 Pa which gives a Young's modulus E of 51509.9 Pa. These values fulfil all the expectations from the composition developed in this study. Chenite et al. [23], on which the composition of the tested hydrogel is based, measured a storage modulus G' between 3000 Pa and 4000 Pa at physiological temperature (37°C). The use of alginate and calcium chloride demonstrates an impressive enhancement of mechanical properties.

4.3.2 Tensile test on hydrogel

In order to evaluate the Young's modulus, a tensile test have been performed on the hydrogel samples. The UniVert CellScale tensile test machine was kindly provided by the SMal laboratory at EPFL. This tensile testing machine allowed us to perform the test with the sample immersed in water at 37°C, thus close to the real situation application (see Appendix 4). However, the samples' shape was rectangular and not the traditional dogbone shape due to the difficulty of casting such shapes with the viscosity of the synthesized hydrogel.

Based on ASTM norme, the deformation rate applied during the test is 100mm/min with load cell of 1N. The results obtained are shown in Figures 8 and 9.

The Young's modulus E is around 40 kPa thanks to linear regression by averaging over the three samples tested. This result is close to the one approximated from rheological tests (≈ 51 kPa). Plus, the hydrogel did not break after the maximal imposed elongation.

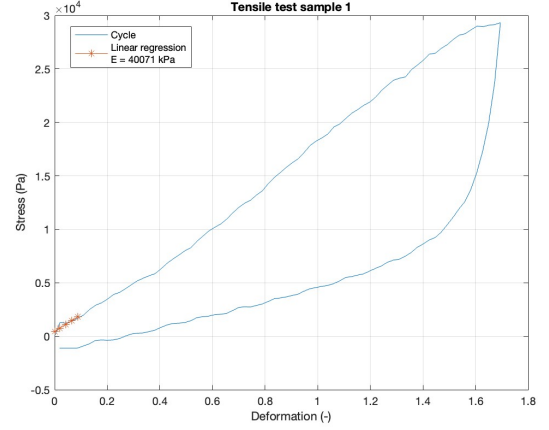


Figure 8: Tensile test to evaluate the Young's modulus.

Thus, it has been chosen to perform multiple cycles in order to determine how the material reacts under cyclic conditions. Consequently, the hydrogel still did not break even after 5 cycles, but it homogeneously deformed plastically (deformed rectangular shape). Cyclic test, as shown in Figure 9, exhibits first an expected decrease of the properties after the first cycle due to (ionic) bonds ruptures. However, the hydrogel keeps suitable properties through multiple cycles which is indispensable for the real application.

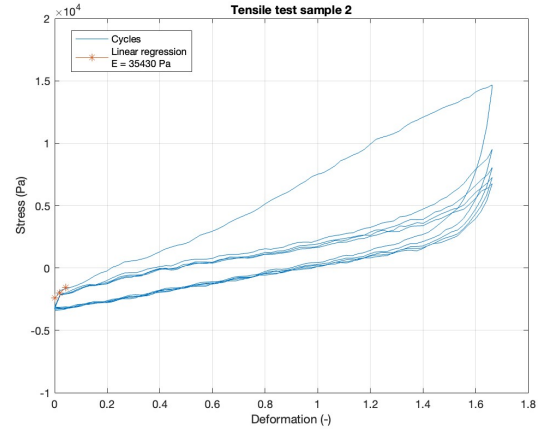


Figure 9: Tensile test to estimate the cyclic behavior.

4.3.3 Injectability test

The injectability has been evaluated visually in the lab by filling a syringe with the solution freshly prepared and then pushing the piston in order to see if the solution were able to flow through the needle. The test was successful using a Microlance needle 20G $\times$ 1 1/2" 0.9 $\times$ 40mm as used during a surgery. The success is demonstrated by some pictures in Appendix 3.

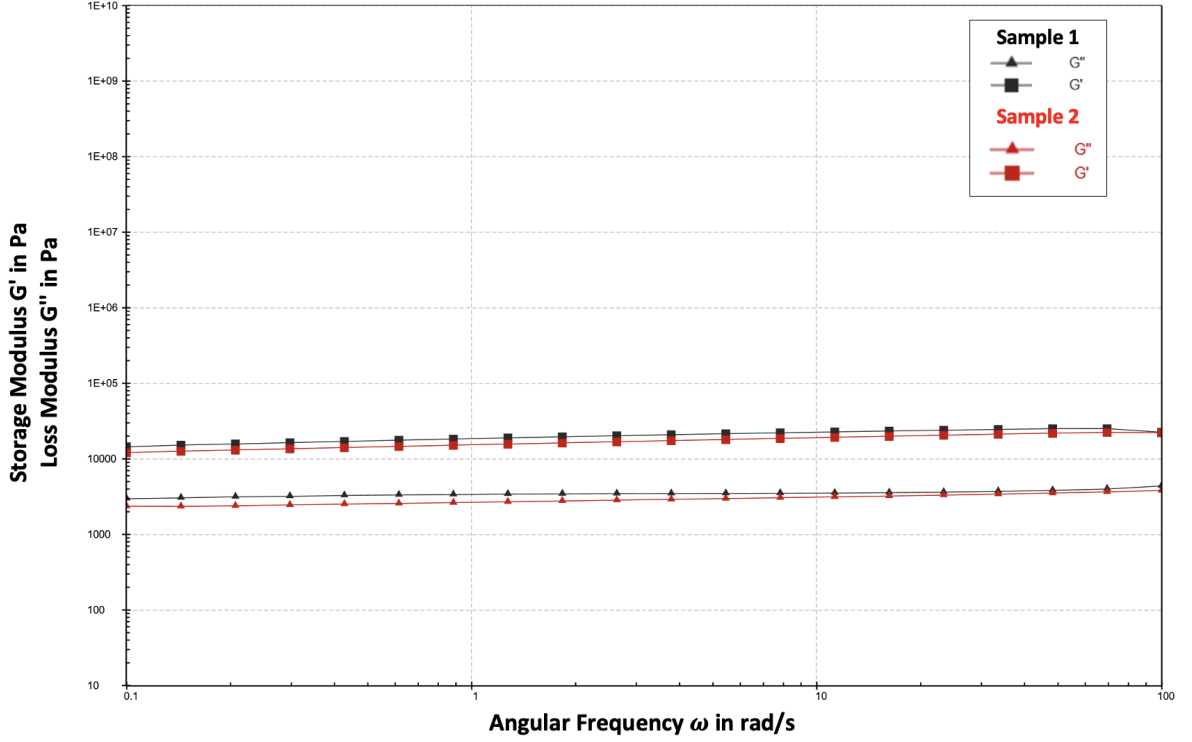


Figure 7: Frequency sweep to evaluate G' and G'' performed at 37 °C.

Finally, Table 10 summaries properties of the created hydrogel.

Table 10: Overview of hydrogel’s properties

Properties	
Young’s Modulus	45 kPa
Deformation at rupture	$\geq 180\%$
Poisson’s ratio	0.35
pH	≈ 6.8
Injectability	
Biocompatibility	

injectable hydrogel using chitosan, glycerophosphate, and alginate as the main components. Our biocompatible hydrogel displayed a Young’s modulus of ≈ 40 kPa, which is more than 5 times higher than the value for the soft palate of a patient suffering from OSA (Haddad et al., 2018) and a simple formulation which doesn’t require difficult technical steps to produce it. The stiffness provided by the hydrogel is a critical property for providing internal support to the soft palate and preventing its displacement. These promising results underscore the feasibility of our proposed procedure and paves the way for further ex-vivo experimentation on palatal tissues.

5 Conclusion

Obstructive Sleep Apnea (OSA) significantly impairs the sleep quality of millions people globally, necessitating improved treatment approaches. Addressing the limitations of current methods, we propose the use of an injectable hydrogel to support the soft palate internally and prevent upper airway obstruction. Our numerical study demonstrates that injecting a stiffening hydrogel into the soft palate can reduce its maximal displacement by 23%, effectively eliminating upper airway obstruction in OSA patients. Expanding upon these findings, we formulated a new

Limitations

While our study presents a promising approach to preventing obstruction in OSA using an injectible hydrogel, there are several areas that require further investigation before progressing to in vivo studies for OSA treatment. In particular, the mechanical properties of the hydrogel may be improved to prevent significant displacement of the palate and enhance overall performance. As the soft palate is in contact with body fluid, it is also important to ensure that the injectable hydrogel do not show excessive swelling or degradation, which may impact its mechanical properties.

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Authors Contributions

Conceptualization: L.E., L.J., A.V. and K.Z.; Methodology: L.E., L.J., A.V. and K.Z.; Literature Review: L.E., L.J., A.V. and K.Z.; Numerical Model: L.E. and K.Z.; Hydrogel Synthesis and Testing: L.J. and A.V.; Writing – Original Draft: L.E., L.J., A.V. and K.Z.; Writing – Review and Editing: L.E., L.J., A.V. and K.Z.

Declaration of Interests

The authors declare no competing interests.

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Appendices

Appendix 1

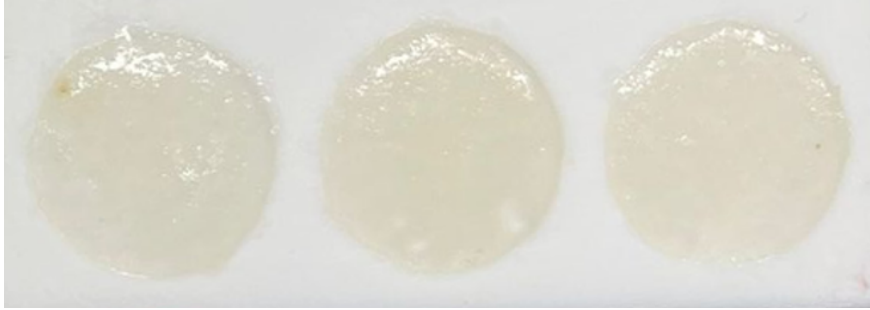


Figure 10: Cast samples for rheological tests

Appendix 2

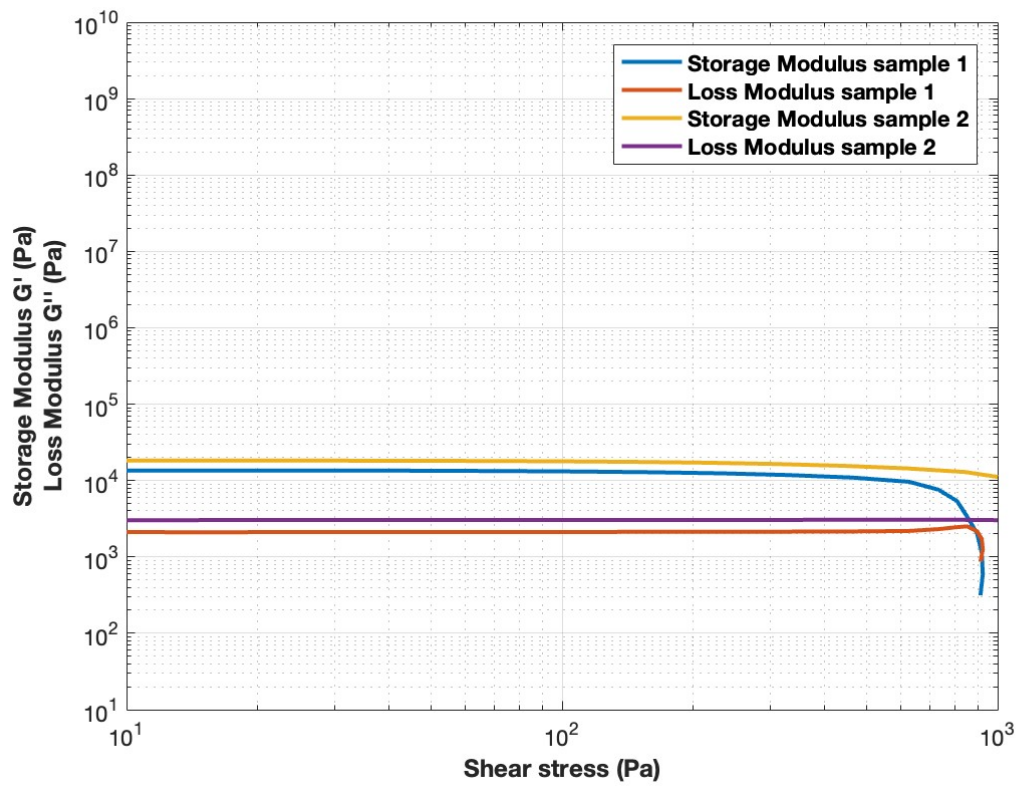


Figure 11: Amplitude sweep of two samples to estimate the amplitude range in the linear viscoelastic region of the hydrogel

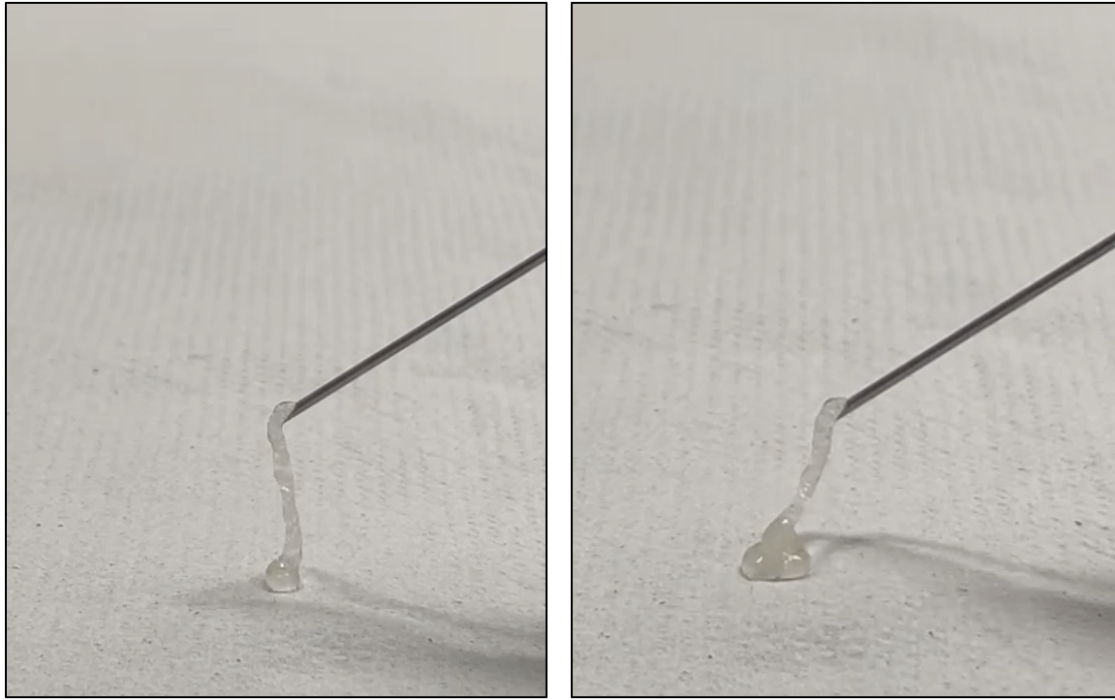
Appendix 3

Figure 12: Proof of injectability with a Microlance needle 20G $\times$ 1 1/2" 0.9 $\times$ 40mm

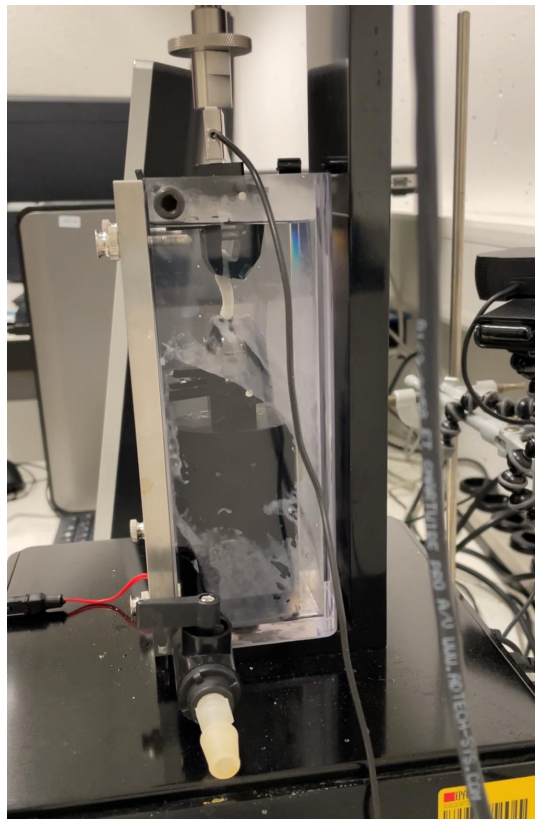
Appendix 4

Figure 13: Set up used to perform a tensile test

Appendix 5

Table 11: Mesh study results: Element number, element number in fluid, computation time, the maximum velocity and the respective relative error between refinement steps

	coarsest	coarser	coarse	normal
Element number	28715	30061	65066	81189
In Fluid	20697	27462	60491	74505
Computation Time [s]	51	82	180	268
$u_{\max}$ [m/s]	0.415	0.923	1.049	1.095
ϵ_{rel}	-	0.5504	0.1201	0.0420

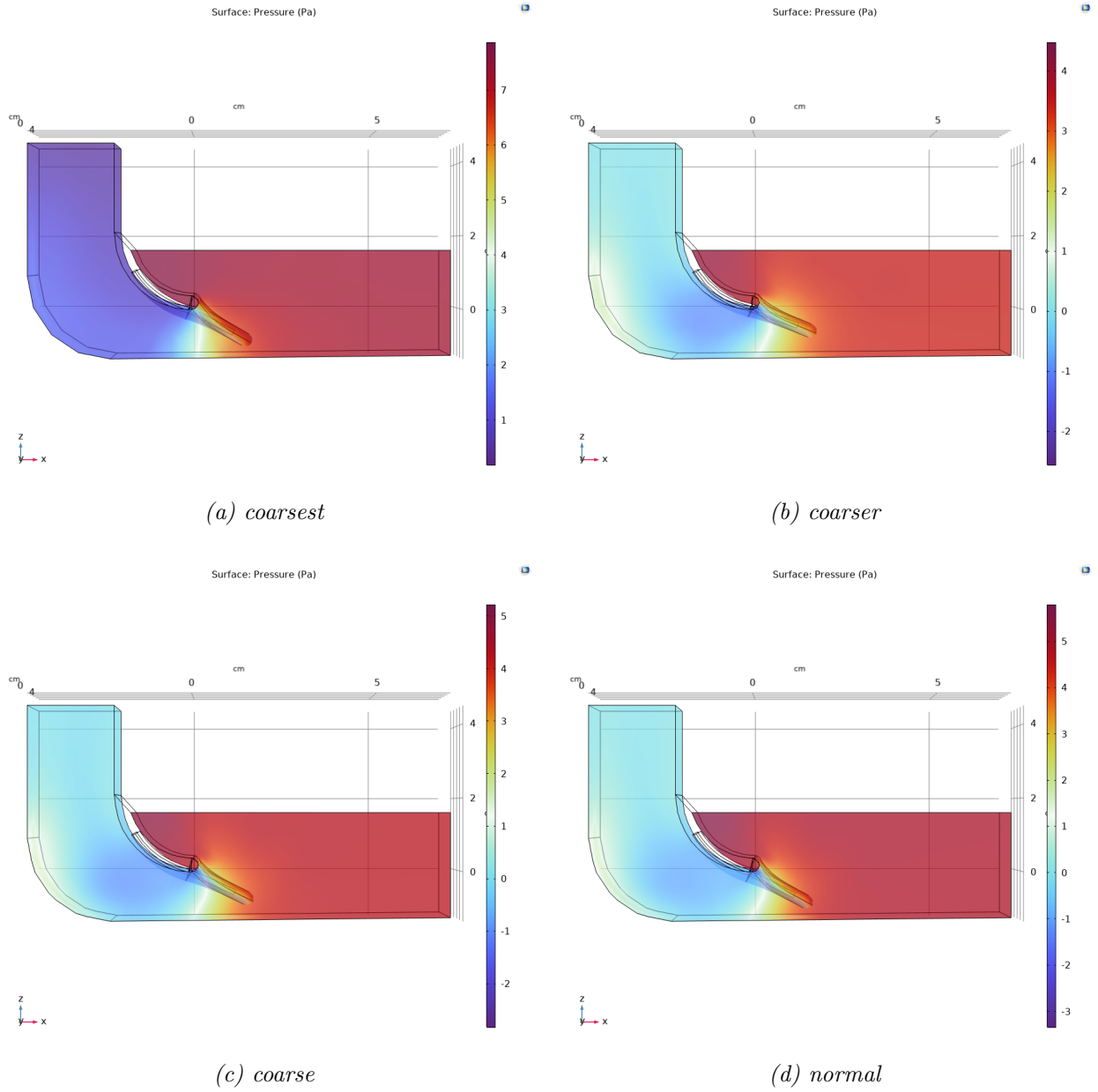


Figure 14: Pressure fields for mesh study cases

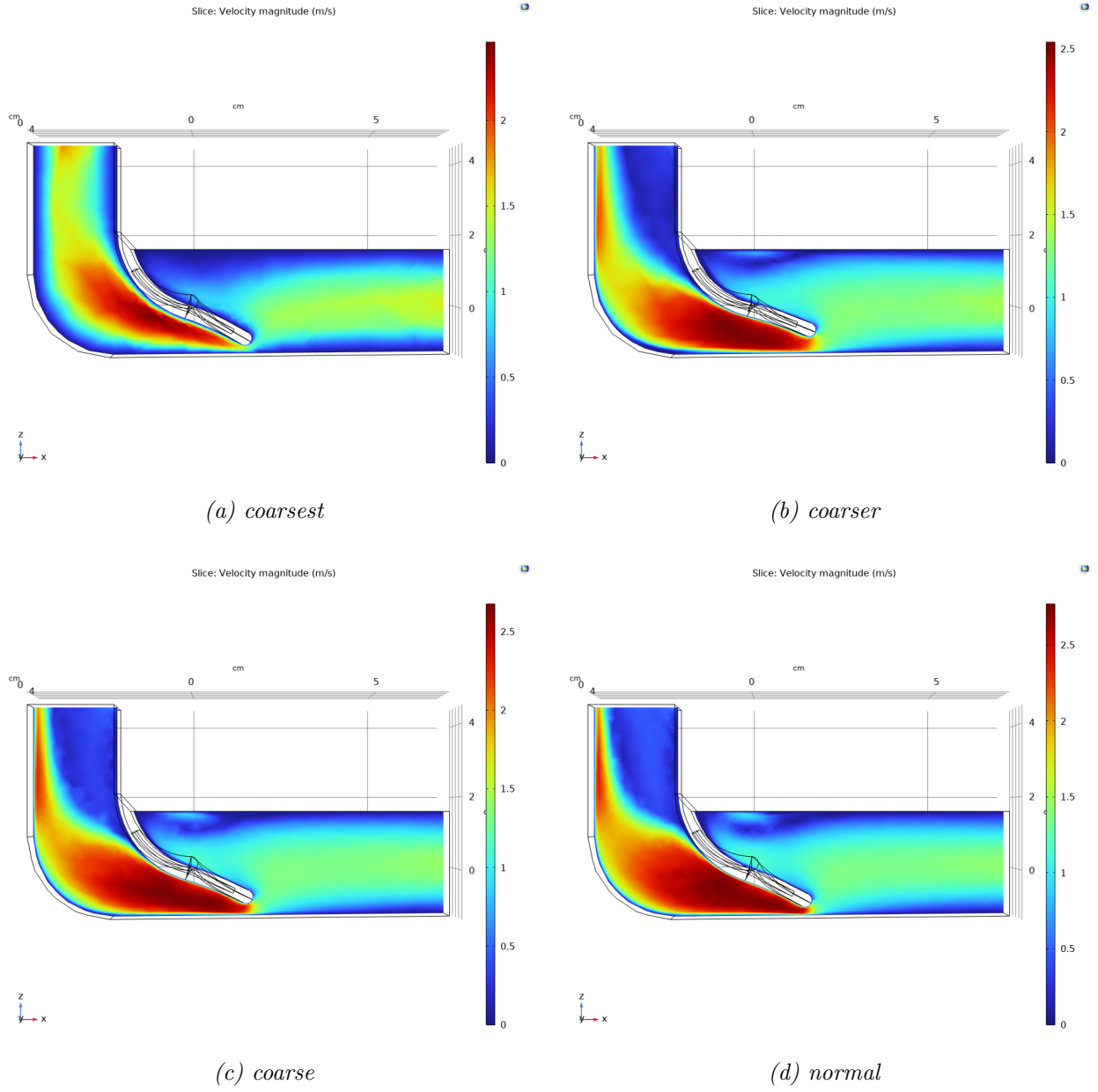


Figure 15: Velocity fields for mesh study cases